# LINA HARCHARRAS

Data Analysis & AI

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## PROFESSIONAL SUMMARY

Computer Science graduate (Big Data Analytics specialization) with hands-on experience across data pipelines, dashboards, machine learning, and AI-powered applications. Comfortable across the analytics stack — Python, SQL, Power BI — with a growing security lens from cloud/network security coursework. Seeking flexible, remote data analyst/scientist roles compatible with graduate studies.

## PROFESSIONAL EXPERIENCE

**Moroccan National Olympic Committee**

*Computer Science Intern — Rabat, Morocco — Jun 2025 – Aug 2025*

- Built an OCR system for Moroccan ID documents using OpenAI and Gemini APIs within a Flask app.
- Integrated a Retrieval-Augmented Generation (RAG) system using Pinecone for semantic search and contextual document retrieval.
- Benchmarked model latency, cost, and accuracy across AI providers.

**ESG Partners (Remote)**

*MIS Intern — Wrocław, Poland — Mar 2025 – May 2025*

- Developed an IT Asset Lifecycle Management System in Python, Streamlit, and Firebase.
- Created interactive dashboards in Power BI for carbon impact analysis.

**AUI School of Science and Engineering**

*Teaching Assistant — Ifrane, Morocco — Jan 2026 – Jun 2026; Sep 2023 – May 2024*

- Graded assignments and exams and supported students in computer science courses, including IoT and Big Data Streaming.

**AUI Center for Learning Excellence**

*Peer Mentor — Ifrane, Morocco — Aug 2025 – Jan 2026*

- Mentored first-year students in academic planning and university integration.

**Mercer University – Rec. Sports & Wellness**

*Sports Official — Macon, GA, USA — Sep 2024 – Dec 2024*

- Officiated intramural soccer, volleyball, and flag football, applying conflict resolution under pressure.

## TECHNICAL SKILLS

- **Programming:** Python, Java, C/C++, SQL
- **Data & AI:** Machine Learning, NLP, scikit-learn, PyTorch, Hugging Face Transformers, FAISS, Qiskit, RAG Systems, OpenAI APIs
- **Data Engineering:** Apache Airflow, dbt, ETL Pipelines, pandas, Big Data Processing
- **Web Development:** Flask, Streamlit, HTML/CSS, REST APIs
- **Database Systems:** PostgreSQL, MySQL, SQLite, Firebase, Data Modeling
- **Networking & Security:** Encryption, Cloud Security, Network Operations, Firewalls
- **Tools & OS:** Power BI, Git, GitHub Actions, Docker, Linux, Windows Server

## LANGUAGES

Arabic (Native), English (C2), French (C1), Spanish (A2), Italian (A1)

## EXTRACURRICULARS

- Captain of the AUI Women's Futsal Team, National University Champions
- Member of Rotaract Humanitarian Club
- Interests: Football, Volleyball, Basketball, Guitar, Music

## EDUCATION

**Al Akhawayn University, Ifrane, Morocco**

*B.S. Computer Science, Minor in Business Administration — GPA: 3.7/4.0*

Relevant Coursework: Algorithms, Database Systems, Data Science, Big Data Analytics, IoT and Big Data Streaming, Advanced and Distributed Programming Paradigms

**Mercer University, USA**

*Exchange Semester — GPA: 4.0/4.0*

## RESEARCH & PROJECTS

**Football Analytics Pipeline & Match Predictor**

*Personal Project — 2026 — Live dashboard*

- Built an automated pipeline ingesting 32,000+ matches from 8 European leagues (CSV + REST API) into PostgreSQL, modeled with tested dbt layers, refreshed daily via GitHub Actions and served on Streamlit.
- Developed an Elo + Poisson model with leakage-free features; a walk-forward backtest on 8,748 matches reached 51.2% accuracy, closing 80% of the RPS gap between a naive baseline and bookmakers.

**Quantum ML for Physiological Stress Classification**

*Research, with K. Qribis — 2026 — Manuscript in preparation*

- Comparing quantum (QSVM, VQC), hybrid, and classical models for stress detection from ECG, EDA, respiration, and EEG signals (WESAD, DREAMER).
- Evaluating subject-independent and cross-dataset generalization.

**Predictive Injury Detection for Athletes (Capstone)**

*Al Akhawayn University — 2026*

- Built a multimodal pipeline fusing wearable IMU, heart rate, EMG, and temperature data, recorded live via Arduino and Bluetooth sensors.
- Benchmarked Random Forest, SVM, and Logistic Regression against a Quantum SVM (Qiskit, 4-qubit ZZ feature map) across 28 configurations with LOFO cross-validation.

**News Article Recommender**

*Course Project, with G. Nafa — 2025*

- Built a semantic recommender with DistilBERT embeddings, FAISS similarity search, and a Streamlit interface, with personalized recommendations from reading history.

**Big Data Analytics Projects**

*Al Akhawayn University — 2025*

- Built near-duplicate detection systems using MinHash and LSH: a plagiarism detection web app (Flask) and an image deduplication pipeline with perceptual hashing, cutting pairwise comparisons by 78%.
- Designed batch/streaming ETL pipelines with Apache Airflow and a PyTorch model to forecast next-day temperature.

**Sales Analytics Dashboard**

*Personal Project — 2026*

- Designed an interactive Power BI dashboard on the Kaggle Superstore dataset, backed by Python/SQLite ETL, to surface sales and margin trends.

**Expense Tracker Web App**

*Personal Project — 2026*

- Built a multi-user expense tracker (Python, Streamlit, SQLite) with budget alerts, Excel export, and AI-generated spending summaries via the OpenAI API.

## CERTIFICATIONS

- **IBM:** Tools for Data Science, Python for AI & Development (2024)
- **U. Minnesota:** Cloud App Security, Cloud Data Security, Cloud Top 10 Risks (2025)
- **Coursera:** Cloud Security, Cybersecurity Essentials (2025)
- **AUI & Internet Society:** Encryption Workshop, Network Operations (2025)
- **Red Crescent:** First Aid Certification (2023–2025)